The first law

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a consistent sign convention in an energy balance.

SUCCESS CRITERIA - I can...

1. State whether heat enters or leaves.
2. Distinguish work done by the system from work done on it.
3. Apply ΔU=Q-W with the stated signs.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “internal energy” and “work” to explain your model or result. Sentence starters are optional.

internal energy: energy stored in a system’s microscopic motion/interactions

work: energy transferred by a force along a displacement

surroundings: everything outside the chosen system

01 Bring back an earlier idea

A 0.3 kg liquid releases 100000 J while condensing at constant temperature.
Find latent heat and signed Q for the sample.

Word help: latent heat = energy per mass needed for a specified phase change

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\Delta U = Q - W_{by}$.
Q is positive into the system; W_{by} is positive when the system does work on its surroundings.

Worked reasoning: A system receives 90 J of heat and does 30 J of work on its surroundings. With ΔU=Q-W, internal energy increases by 60 J.

Your turn: Draw labeled arrows crossing a system boundary and mark the 60 J stored increase.

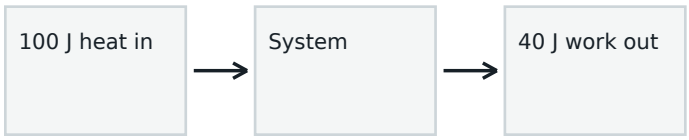
Sentence starter: The … in my model represents … . Its label shows … .

Marzano check - circle: ① ② ③

03 Represent and interpret

A system absorbs 100 J of heat and does 40 J of work.
Draw the energy arrows and find its internal-energy change.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

04 Solve and justify

A gas loses 50 J of heat while 60 J of work is done on it.
Use the reference sign convention to calculate ΔU.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

Why does the phrase “work is 60 J” need a direction or sign convention?

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Choose a method

A 4 kg cart at 12 m/s hits an identical cart at rest. Final velocities are 5 and 7 m/s.
Check momentum and energy; classify the collision.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

07 Exit check

A process hasΔU=-15 J and W_{by}=25 J.
Find Q and say whether heat enters or leaves.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A system loses 20 J of heat and has 50 J of work done on it. Using work-by-system as positive, find Q, W and ΔU. Then reverse both transfers.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3